MILITARY SPECIFICATION

MOUNTER-DEMOUNTER,

PNEUMATIC TIRE, STATIONARY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types (capacities) of stationary, power-driven, pneumatic tire mounter-demounters.

1.2 Classification. Mounter-demounters shall be of the following types, as specified (see 6.2):

Type I - tire capacity as specified in 3.5.2.1.
Type II - tire capacity as specified in 3.5.2.2.
Type III - tire capacity as specified in 3.5.2.3.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CC-M-1807	-	Motor, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)
PPP-B-621	-	Box, Wood, Nailed and Lock Corner
PPP-B-636	-	Box, Shipping, Fiberboard
PPP-C-843	-	Cushioning Material, Cellulosic
PPP-T-60	-	Tape, Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Materiel Readiness Command, ATTN: DRSAR-LEE-S, Rock Island, IL 61299 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	-	Preservation, Methods of
MIL-B-121	-	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-C-5501	-	Cap and Plug, Protective, Dust and Moisture Seal
MIL-H-6083	-	Hydraulic Fluid, Petroleum Base, For Preservation and Operation
MIL-E-16298	-	Electric Machines Having Rotating Parts and Associated Repair Parts; Packaging of
MIL-C-19834	-	Plate, Identification, Metal Foil, Adhesive Backed
MIL-C-52950	-	Crate, Wood, Open and Covered

STANDARDS

FEDERAL

FED-STD-H28	-	Screw Thread Standards for Federal Services

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of US Military Property
MIL-STD-889	-	Dissimilar Metals
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods
MS16658	-	Connector, Plug and Duplex Receptacle, Electrical, Power, 2-Pole, 3-Wire, Grounding Type, 15-Ampere, 125-Volts

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

7353294	-	Standard Military Wheels (Table I)
7353672	-	Standard Military Wheels (Table I)
7353765	-	Standard Military Wheels (Table I)
7355430	-	Standard Military Wheels (Table I)
7355436	-	Standard Military Wheels (Table I)
7355439	-	Standard Military Wheels (Table I)
7387807	-	Standard Military Wheels (Table I)
7388452	-	Standard Military Wheels (Table I)
7388820	-	Standard Military Wheels (Table I)
7389618	-	Standard Military Wheels (Table I)
7389621	-	Standard Military Wheels (Table I)
FCPR100057	-	Tire, HET
KCPR100058	-	Wheel, HET
FCPR103780	-	Tire, HET
KCPR104507	-	Wheel, HET
KCPR109474	-	Wheel, HET

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

201.02 - Tooth Proportions for Coarse-Pitch Involute Spur Gears
390.03 - Gear Classification Manual
600.01 - Metric Gears

(Application for copies should be addressed to American Gear Manufacturer's Association, One Thomas Circle, Washington, DC 20005.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C1 - National Electrical Code
ANSI/ASTM E 380 - Metric Practice

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA)

Electrical Standards

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 155 East 44th Street, New York, NY 10017.)

INTERNATIONAL STANDARDS ORGANIZATION (ISO)

ISO Standard 1000 - International System of Units (SI)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one complete mounter-demounter for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The mounter-demounter shall be actuated by electric over hydraulic or pneumatic power and shall be designed to mount and demount tires from the rims of passenger cars, military vehicles, and commercial light and heavy trucks. The mounter-demounter shall mount and demount tires to and from rims of the sizes specified herein. The units and their attachments shall be designed with substantial strength to break tire beads loose and perform the stated mount and demount operations. The mounter-demounter shall be designed so that raising of the tire and wheel assembly above floor level by manual means is not required to mount the assembly on the wheel-holding unit.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the tire support industry. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Reclaimed materials. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.4 Construction. The mounter-demounter shall be complete so that, when installed or connected to its specified source of power, it can be used for any operation for which designed. Construction shall be free from any characteristic or defect that would prevent the mounter-demounter from performing any of the requirements specified herein.

3.4.1 Castings and forgings. All castings shall be free of blow holes, porosity, hard spots, cracks, and other surface defects. Forgings shall be free from scale, cold shuts, mismatching, and other surface defects. Strength and other essential properties of the castings and forgings shall meet the performance requirements specified herein.

3.4.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations, however, shall not be employed as repair measures for defective parts.

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3.4.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked, or otherwise deformed to the extent that the parts cannot be readily disassembled and reassembled.

3.4.4 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering. Rosin base fluxes only shall be used in soldering operations. Connections to screw-type terminals shall be mechanically secured to prevent loss of tightness.

3.4.5 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.4.6 Gears. All gears and pinions used in the mounter-demounter or its components shall be fabricated in either the English (USA) system or the Metric (SI) system of measurement. English gears shall conform to AGMA 390.03. Spur gears shall conform to AGMA 201.02. Metric gears shall comply with the International System of Units (SI), International Standards Organization (ISO) Standard 1000, American National Standard ANSI/ASTM E 380, and American Gear Manufacturers Association 600.01.

3.4.7 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible. Lubrication reservoirs, if applicable, shall have means for checking levels.

3.4.8 Safety devices. Each mounter-demounter shall be furnished with safety devices. Parts which are hazardous to the operator shall be guarded. Protection against electrical shock shall be provided. When specified (see 6.2), special safety devices shall be provided as required by the procuring activity. Sharp corners and edges that present hazards shall be removed or shielded.

3.4.9 Interchangeability. All replaceable parts shall be designed and manufactured to such standards as shall permit replacement and adjustment without the need for modification of the parts or the mounter-demounter.

3.4.10 Surfaces. All surfaces of castings, forgings, molded parts, stampings, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

3.5 Performance. The mounter-demounter shall incorporate facilities for holding, removing, and installing pneumatic tires specified in 3.5.2 without damage to tire or wheel. Soap and water or any applicable rubber lubricating solution which shall not injure the tires may be used as a lubricant during the mounting and demounting operation.

NOTE: Damage to tire beads and sidewalls, such as gouging or tearing of the rubber, shall be cause for rejection. Damage to rim or wheel flanges and bead seats constituting permanent deformation of the metal beyond its initial manufactured state shall be cause for rejection.

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3.5.1 Power source. Electric motors shall operate from 120 volt or 220/230 volt service. Mechanisms utilizing compressed air as a power source shall function and perform with 100 pounds per square inch gage (psig) air pressure, as specified herein. Mechanisms that function and perform on less than 100 psig are also acceptable.

3.5.2 Capacities. The mounter-demounters shall mount and demount tires to and from rims in the capacities specified in 3.5.2.1 through 3.5.2.3.

3.5.2.1 Type I tire capacity. The type I mounter-demounter shall mount and demount, from drop-center rims, tire sizes 7.50-10 through 8.00-17.5 (tires having a maximum 8-ply rating) and shall mount or demount in a horizontal or vertical plane.

3.5.2.2 Type II tire capacity. The type II mounter-demounter shall mount and demount, from flat-type rims, tire sizes 9.00-16 through 14.00-24, Table I, sizes 2 through 11. The type II mounter-demounter shall also demount, from drop-center rims, tire size 7.00-16, Table I, size 1.

TABLE I. Wheels for Type II mounter-demounter.

Size No.	RIM Dia.	RIM Size	WHEEL Hub Bore	WHEEL Offset	Approx. wt. lbs. (complete)	Dwg. Number	Tire Application
1	16	4.50E0	4.315	3/8	19.5	7387807	7.00-16 Single 6 Ply
2	16	6.50H	4 31/32	4	44.0	7388452	9.00-16 Single 8 Ply
3	16	6.50H	4 31/32	5 7/8	46.0	7355436	9.00-16 Dual 8 Ply
4	15	7.5	6 7/16	6 1/8	65.5	7355430	9.00-15, 10.10-15 Dual 12 & 14 Ply
5	20	7.5	6 7/16	5 1/8	88.0	7389618	11.00-20 Single 12.50-20 Single 12 Ply
6	20	7.5	6 7/16	6 1/4	85.0	7389621	9.00-20 Dual 8, 10, & 12 Ply
7	20	7.5	8.725	6 3/4	92.0	7388820	10.00-20, 11.00-20 Dual 12 Ply 11.00-20 Single (Special)
8	20	7.5	8.725	7 1/8	89.0	7353294	12.00-20 Dual 14 Ply
9	20	10.0	8.725	5 3/8	140.0	7353765	14.00-20 Single 12, 18, & 20 Ply
10	20	10.0	8.725	8 7/8	140.0	7353672	14.00-20 Dual 12, 18, & 20 Ply
11	24	10.0	10.671	8 7/8	171.0	7355439	14.00-24 Dual 18 & 20 Ply

3.5.2.3 Type III tire capacity. The type III mounter-demounter shall be capable of mounting and demounting both tube and tubeless-type pneumatic tires 7.00-15 through 14.00-24 to 18-22.5, 2-ply through 22-ply, from commercial wheels and from the standard military wheels in Table I. The mounter-demounter shall also mount and demount tires from the following wheel/rim/tire sizes specified in Table II.

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TABLE II. Wheels for Type III mounter-demounter.

Drop Center Rim			Flat Base Rim	
7.00-15	8 ply		7.00-15	8 ply
8-19.5	8 ply		11.00-15	16 ply
9.50R-16.5D	8 ply			
11.00-18	6 ply			
15.00-19.5	14 ply			
16.5-19.5	16 ply			
18.00-22.5	22 ply			

In addition to the above, the type III mounter-demounter shall be capable of mounting and demounting Heavy Equipment (HEI) Tires/Wheels identified as follows:

Tire - Goodyear Super Single, 18-22.5, 22 ply (Army Drawing FCPR10 3780) used with:
Wheel (Army Drawing KCPR109474)
Tire - Goodyear Super Single, 15-19.5, 14 ply (Army Drawing FCPR100057) used with:
Wheel (Army Drawing KCPR104507)
Wheel (Army Drawing KCPR100058)
Tire - 15x19.5, 14 ply, used with:
Wheel Army Drawing KCPR104507)

3.5.3 Operational requirements. The operation of each mounter-demounter shall require not more than one operator.

3.6 Types.

3.6.1 Details of components for type I. The type I tire mounter-demounter shall consist of the following principal components:

3.6.1.1 Pedestal. The pedestal shall be made of metal capable of withstanding the operations specified herein. The pedestal shall support the wheel so that the tire mounting-demounting operations are performed with the wheel positioned in a horizontal or vertical plane. The pedestal shall provide a method for securing the mounter-demounter to the floor if required.

3.6.1.2 Wheel-retaining device and rotating unit. A wheel-retaining device shall be provided to locate and retain the wheel in position on the pedestal for mounting and demounting. All wheels specified in 3.5.2.1 shall be readily retained and positioned on the device, using adapters if necessary, for mounting or retaining.

3.6.1.3 Bead breaker. The bead breaker shall free both beads of the tire from the rim while the wheel is secured to the pedestal. It shall not be necessary to remove and replace the wheel on the pedestal to perform this operation. The bead breaker shall be adjustable to accommodate all tires used on wheels specified in 3.5.2.1. The bead engaging surfaces of the bead breaker shall be designed so as not to cause damage to the rim, the tire bead, the inside of the tire, or the tire tube as specified in the note under 3.5.

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3.6.1.4 Pneumatic or hydraulic pressure unit(s). Pressure unit(s) shall be furnished to provide the operating force for the bead breaker unit(s) specified herein. The pressure unit(s) shall embody a hand or power-actuated pump(s) and power ram(s). The control(s) shall be located to permit the operator to control the bead-breaker unit(s) while applying pneumatic or hydraulic pressure. The pressure unit(s) shall be capable of delivering a force that shall break the beads of the tires specified in 3.6.2.1 and 3.5.3, under the conditions specified in 3.5 and 3.5.1, by actuating the pump control(s). Pressure shall be maintained for not less than two minutes at a force of at least five psig with the units showing no evidence of leakage. Air hoses shall be of proper length and design such that no kinks in the hoses shall occur during normal installation and operation.

3.6.1.5 Mounting and demounting tools. The tire mounter-demounter shall have a tool or device to facilitate removal of the tire from the wheel and installation of the tire on the wheel, if such a tool is required to perform the operation. If one is required to perform the operation, a separate tool may be provided for each operation. For the demounting operation, the tool shall separate each tire bead from the rim. The tool, or tools, used for each operation shall be designed with bead-engaging surfaces so that no damage is incurred in the tire bead, the inside of the tire, or the tire tube. The tool or tools shall be designed to accommodate all tires used on the wheels specified in 3.5.2.1.

3.6.2 Detail of components for type II. The type II mounter-demounter shall consist of the following principal components:

3.6.2.1 Base. The base shall be constructed of steel capable of supporting the mounter-demounter during the operations specified herein. The base shall incorporate a method for securing the mounter-demounter to the floor to retain the mounter-demounter in a stationary position during the tire mounting and demounting operations.

3.6.2.2 Wheel-holding unit. The wheel-holding unit shall embody a retaining device and locking mechanism or adaptor(s). The wheel-holding unit shall mount and retain the wheel and tire assemblies, within the tire capacities specified for type II (3.5.2.2), on the mounter-demounter. Rotary-type mounter-demounters shall be capable of demounting the tire in only one direction of rotation. Wheel-holding units shall grip and retain the wheel by positive action such that the wheel shall not be loosened without deliberate positive action by the operator. The wheel shall not be released if, or when, the rotation is reversed and the tools become jammed between the tire and wheel.

3.6.2.3 Bead-breaker unit(s). The mounter-demounter shall incorporate a bead-breaker unit(s) for breaking the outer and inner tire beads away from the wheel rim. The bead-breaker unit(s) shall completely remove tires on drop center rims and also remove tires on flat-type rims after removal of the locking ring. This shall be accomplished without the necessity of repositioning the wheel assembly on the wheel holding and rotating device. The bead-breaker unit(s) shall not interfere with the mounting of the wheel and tire assembly on the wheel-holding unit and shall incorporate a positive adjustment, with controls, to securely position the bead breaker unit(s) to accommodate the tire capacities specified in 3.5.2.2.

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3.6.2.4 Pressure unit(s). Pressure unit(s) shall be furnished to provide the operating force for the bead-breaker unit(s) specified herein. The pressure unit(s) shall embody a hand- or power-actuated pump(s) and power ram(s). The hand levers shall be located to permit the operator to control the bead-breaker unit(s) while applying the pneumatic or hydraulic pressure. The pressure unit(s) shall be capable of delivering a force that shall break the beads of the tires specified in 3.5.2.2 and 3.5.3, under the conditions specified in 3.5 and 3.5.1, by actuating the pump control. Pressure shall be maintained for not less than two minutes at a force of at least five psig with the unit showing no evidence of leakage. The pressure unit(s) shall include provisions to automatically, or by pump action, return the power ram(s) when pressure is released.

3.6.2.5 Wheel-lifting unit. The mounter-demounter shall incorporate a powered lifting unit capable of lifting the wheel and tire assemblies specified in 3.5.2.2 from floor level to the demounting position and lowering the wheel and rim assembly to floor level, after removal of the tire, in such a manner that the wheel is not driven into the floor. The lifting unit shall be operated by a foot or hand control.

3.6.2.6 Motor housing. The mounter-demounter shall incorporate a housing to enclose the drive mechanism. The housing shall be fabricated of sheet steel of not less than Manufacturers' Standard Gage No. 18. The housing shall be removable or shall include removable panels to afford access to the motor and drive.

3.6.3 Detail of components for type III. The type III tire mounter-demounter shall consist of the following principal components.

3.6.3.1 Base. The base shall be of reinforced welded construction or a casting, with means for leveling and bolting to the floor.

3.6.3.2 Wheel-holding unit. The wheel-holding unit shall embody a retaining device and a locking mechanism. The wheel-holding unit shall hold the tire and wheel assemblies specified in 3.5.2.3 on the mounter-demounter. Tube-type tires and wheels may be held on mounting spools and adaptors.

3.6.3.3 Bead-breaker unit(s). The mounter-demounter shall incorporate bead-breaker unit(s) for breaking the bead away from the flange, with the lock ring side of the rim out, and depressing and holding the tire until the lock ring is removed.

3.6.3.4 Pressure unit(s). A pressure unit(s), capable of delivering a force that shall break the beads of the tires specified in 3.5.2.3, including Table II, and the wheels and tires following Table II, shall be provided for the bead breaker. Pressure shall be maintained for not less than two minutes. The unit(s) shall show no evidence of leakage.

3.6.3.5 Motor housing. The mounter-demounter shall incorporate a housing to enclose the drive mechanism. The housing shall be fabricated of sheet steel of not less than Manufacturers' Standard Gage No. 18. The housing shall be removable or shall include removable panels to afford access to the motor and drive.

3.6.3.6 Wheel-lifting unit. The type III mounter-demounter shall incorporate a power lifting unit capable of lifting the wheel and tire assemblies specified in 3.5.2.3 from floor level to the demounting position and lowering the

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wheel and rim assembly to floor level, after removal of the tire, in such a manner that the wheel is not driven into the floor. The lifting unit shall be operated by a foot or hand control.

3.7 Electrical equipment. The electrical components for all three types of the mounter-demounter shall conform to applicable NEMA Standards, except as specified herein. The mounter-demounter shall incorporate, but not be limited to the following components (3.7.1 thru 3.7.3).

3.7.1 Motor. The motors for driving the mounter-demounter specified in 3.5.1 shall be of sufficient horsepower to meet the requirements specified herein. Unless otherwise specified (see 6.2), the drive motor shall be an induction squirrel-cage type of a continuous-duty classification, with an ambient temperature rating of 50° centigrade in conformance with CC-M-1807, with Grade B insulation or better. Unless otherwise specified (see 6.2), the motor shall operate on 120 volts, 60 hertz (Hz), single-phase current.

3.7.2 Motor-control switch. A hand-operated or foot-controlled floor switch shall be provided on the operator's side of the machine to permit control of the motor while positioning the bead-breaker unit(s) and applying the pneumatic or hydraulic pressure. An emergency shut-off switch, either hand or foot operated, shall be provided within reach of the mounted-wheel assembly.

3.7.3 Power supply cord. Unless otherwise specified (see 6.2), the motor shall be furnished with not less than eight feet of three-conductor flexible electric cord, No. 12 AWG, type SO heavy duty, in accordance with ANSI C1 for motors wired for 120 volts. The cord shall be provided with a three-prong, polarized connector plug of either the grip-cord type or molded-on type, conforming to MS16658.

3.7.4 Electrical protection. The mounter-demounter's electrical circuits shall be protected to prevent possible damage during current overload or short.

3.8 Painting. Unless otherwise specified (see 6.2), painting and finishing shall be in accordance with standard commercial practice provided the following minimum criteria are met or exceeded: all surfaces to be painted shall be clean and free of all foreign matter detrimental to painting and at least one coat of primer and one coat of enamel, or equivalent, applied.

3.9 Identification marking. Unless otherwise specified (see 6.2), a corrosion-resistant metal name plate in accordance with MIL-P-19834 shall be permanently and legibly marked in accordance with MIL-STD-130, including serial number and National Stock Number, and shall be securely attached to each mounter-demounter.

3.10 Workmanship. Standards of workmanship shall assure that the mounter-demounter shall have the strength, safety, stability, and efficient operating characteristics found in standard commercial units and as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the

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performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of (a) through (c) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Tests (see 4.6).
(c) Packaging inspection (see 4.7).

4.4.1 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with MIL-STD-105 and the following:

EXAMINATION/TESTS	INSPECTION LEVEL	AQL
Examination	General Level II	1.0
Tests	General Level II	1.0
Packaging inspection	Special Level S-4	2.5

4.5 Product examiantion. Each complete mounter-demounter shall be examined to determine compliance with the requirements of Section 3. As a minimum, the following characteristics shall be examined:

CHARACTERISTICS	REQUIREMENT PARAGRAPH
Design defects	3.2
Construction defects	3.4 thru 3.4.10
Interchangeability	See note below.
Components not as specified	3.6.1 thru 3.6.3.6
Electrical equipment not as specified	3.7 thru 3.7.4

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Painting not as specified	3.8
Product marking not as specified	3.9
Workmanship defects	3.11

NOTE: Vendor certification shall be required for all components referenced to product or process specifications and standards. Certification of conformance shall also be required to indicate compliance with interchangeability requirements (see 3.4.9).

4.6 Tests.

4.6.1 Performance test. Unless otherwise specified (see 6.2), one tire for each of the smallest and largest rims covered under the contract or order, of the types specified in 3.5.2.1, 3.5.2.2, or 3.5.2.3, shall be examined and all defects and irregularities noted. If available, these tires should be obtained from vehicles that have been in service in order that conditions will be encountered similar to those found under actual field conditions. The tires shall be deflated and positioned on the mounter-demounter and removed from their rims in accordance with the manufacturer's instructions. The mounter-demounter shall be checked during the mounting and demounting operations, and the tires shall be examined on conclusion of the operations to determine compliance with the following:

(a) Performance requirements specified in 3.5.
(b) Air pressure and electric supply requirements specified in 3.5.1.
(c) Tire capacity range specified in 3.5.2.
(d) Wheel retaining and rotating shaft, retaining adapter, and locking mechanism operation specified in 3.6.1.2, 3.6.2.2, or 3.6.3.2, as applicable.
(e) Outer and inner bead-breaker operation specified in 3.6.1.3, 3.6.2.3, or 3.6.3.3, as applicable.
(f) Operation of pressure unit(s) as specified in 3.6.1.4, 3.6.2.4, or 3.6.3.4, as applicable.
(g) Wheel-lifting unit operation specified in 3.6.2.5 or 3.6.3.6, as applicable.
(h) Motor-control switch operation specified in 3.7.2.

4.6.2 Load sustaining test. The mounter-demounter pressure unit ram(s), in their pressure exerting position, shall be subjected to their rated bead breaking pressure for a period of two minutes. The units shall deliver the supplier's rated design bead-breaking pressure and show no evidence of leakage in compliance with 3.6.1.4, 3.6.2.4, and 3.6.3.4, as applicable.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging of the mounter-demounter to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Disassembly. The mounter-demounter may be disassembled to the extent necessary to assure that all surfaces requiring a preservative are accessible for processing, to assure minimum cube for packing, and to permit reassembly without special skills or tools. Disassembly, when applicable, shall be as specified in MIL-STD-1186.

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5.2 Preservation. Preservation shall be Level A or Industrial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. The mounter-demounter shall be cleaned and dried in accordance with any applicable process and procedure of MIL-P-116.

5.2.1.2 Preservative application. Immediately after cleaning and drying, apply a coat of type P-2 or type P-11 preservative to all exterior, critical, ferrous metal surfaces of the mounter-demounter and accessories. Exterior noncritical, ferrous metal surfaces which have not been painted or plated shall be coated with type P-1 or type P-19 preservative. Interior surfaces of the air motor shall be coated with a preservative oil conforming to type P-10, grade 30. The oil shall be applied by injection into the air stream while operating the motor until the preservative appears at the exhaust port. The hydraulic system shall be flushed with a preservative oil conforming to MIL-H-6083. After flushing, the system shall be thoroughly drained. In the event the hydraulic preservative oil is not compatible with the supplier's unit, it is the responsibility of the supplier to contact the procuring agency for the selection of an alternate preservative. P-type preservatives specified shall conform to MIL-P-116.

5.2.1.3 Unit packs. The method of unit protection shall conform to method I of MIL-P-116. Preserved critical, ferrous metal surfaces shall be wrapped with barrier material conforming to MIL-B-121, type II, grade A, class optional. The hydraulic ram shall be secured in a retracted position. Air inlets and outlets shall be sealed with plastic caps or plugs conforming to MIL-C-5501 or tape conforming to PPP-T-60, class IV. The electric motor shall be packaged in accordance with level A requirements of MIL-E-16298 and the applicable subparagraph for assembled machines. The pressure gage, as applicable, shall be covered with one inch thick uncompressed cushioning material conforming to PPP-C-843, type II. Secure cushioning and barrier material in place with tape conforming to PPP-T-60, type optional. Technical data issued with the unit shall be packaged in accordance with method IC-1 of MIL-P-116. Coil the power supply cord to a minimum safe diameter and secure to the basic unit with cotton tape or twine. The connector plug shall be protected in accordance with method III of MIL-P-116.

5.2.1.4 Consolidated package. Consolidate the packaged tools, technical data, and all small disassembled components and accessories. Package in a snug-fitting fiberboard box conforming to PPP-B-636, class weather-resistant. Cushioning material specified in 5.2.1.3 shall be employed to restrict movement within the box and to protect parts that are vulnerable to damage.

5.2.2 Industrial. Industrial preservation shall be as specified in ASTM D3951-82.

5.3 Packing. Packing shall be Level A or Industrial, as specified (see 6.2).

5.3.1 Level A. Each mounter-demounter shall be packed in a box conforming to PPP-B-621, class 2, or a crate conforming to MIL-C-52950, type V, style B, covered. The selection of the container shall be governed by the limitations of the container specifications. The consolidated package (see 5.2.1.4) shall be

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placed inside the shipping container and secured to restrict movement. The contents of an open crate shall be covered with a waterproof barrier shroud. Anchoring, shrouding, blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186.

5.3.2 Industrial. Industrial packing shall be as specified in ASTM D3951-82.

5.4 Marking. Marking shall be Level A or Industrial, as specified (see 6.2).

5.4.1 Level A. Level A marking shall be as specified in MIL-STD-129.

5.4.2 Industrial. Industrial marking shall be as specified in ASTM D3951-82.

6. NOTES

6.1 Intended use. The mounter-demounter is intended for use in base shops, field service, and maintenance shops to facilitate the changing of tires on passenger cars, military vehicles, and light and heavy commercial trucks and trailers.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) First article when required (see 3.1).
(d) Special safety devices when required (see 3.4.8).
(e) Motors if different and current, voltage, Hz, and phase if different (see 3.7.1).
(f) Power supply cord if different (see 3.7.3).
(g) Painting if different (see 3.8).
(h) Identification marking if different (see 3.9).
(i) First article inspection if different (see 4.3).
(j) Quality conformance inspection if different (see 4.4).
(k) Sampling if different (see 4.4.1).
(l) Performance test if different (see 4.6.1).
(m) Level of preservation, packaging, packing and marking required (see 5.2, 5.3, and 5.4).

6.3 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

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6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - AL
Navy - YD
Air Force - 99

Preparing activity:

Army - AL

Project No. 4910-0573

Review activities:

Army - AT
Navy - None
Air Force - None

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